

Countable Additivity is Not First-Order Axiomatizable

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Abstract

In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes the models in which the charge is σ -additive. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof is a single ultraproduct argument via Łoś's theorem: Dirac masses on $\mathcal{P}(\mathbb{N})$ are individually σ -additive, but the charge induced on the diagonal copy of $\mathcal{P}(\mathbb{N})$ in their ultraproduct is the $\{0, 1\}$ -valued ultrafilter charge, which is purely finitely additive — contradicting Łoś's theorem if any first-order axiomatization existed.

A property of structures is first-order axiomatizable in a language L if there exists a set of L -sentences whose models are exactly the structures satisfying it. Łoś's theorem gives the key diagnostic: if a property fails in an ultraproduct of structures that individually satisfy it, no first-order axiomatization exists.

Countable additivity distinguishes σ -additive measures from finitely additive charges. That it cannot be expressed by a single first-order sentence is clear: the condition quantifies over countable sequences, unavailable in first-order logic. Its non-first-order character is treated as background knowledge in the probability logic literature [2, 3, 1], but a short self-contained ultraproduct proof in the Boolean-algebra-with-charge setting — showing non-axiomatizability by *any* set of sentences, not merely a single formula — seems not to have been recorded.

We provide one here.

Let $\mathcal{L}_{\text{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0, 1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\text{BA},\mu}$ is a Boolean algebra B together with a $[0, 1]$ -valued finitely additive probability μ , encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0, 1]$ avoids introducing a second numeric sort.

Definition 1 (First-order theory of finitely additive probabilities). Let T_{fa} be the $\mathcal{L}_{\text{BA},\mu}$ -theory axiomatizing:

- (i) B is a Boolean algebra;
- (ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;
- (iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;

(iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \Rightarrow M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The theory T_{fa} may be supplemented with rational-cut coherence axioms (monotonicity of the predicates in q , and $M_{\leq 1}(x)$, $M_{\geq 0}(x)$ for all x) to ensure that the predicates determine a unique value $\mu(x) \in [0, 1]$ for each element; the proof below does not depend on whether these are included.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\text{BA}, \mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n \in \mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\text{BA}, \mu}$.

Proposition 2 (σ -additivity is not first-order axiomatizable). *There is no first-order $\mathcal{L}_{\text{BA}, \mu}$ -theory $T \supseteq T_{\text{fa}}$ such that, for every $(B, \mu) \models T_{\text{fa}}$,*

$$(B, \mu) \models T \iff \mu \text{ is } \sigma\text{-additive}.$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let δ_n be the Dirac probability at $n \in \mathbb{N}$ on the σ -algebra $\mathcal{P}(\mathbb{N})$. Each $(\mathcal{P}(\mathbb{N}), \delta_n)$ is a model of T_{fa} in which δ_n is σ -additive. Suppose for contradiction that a theory T as described exists; then each $(\mathcal{P}(\mathbb{N}), \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. By Łoś's theorem [4], the ultraproduct $\prod_{\mathcal{U}}(\mathcal{P}(\mathbb{N}), \delta_n)$ also models T . The charge induced on the diagonal copy of $\mathcal{P}(\mathbb{N})$ is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{n \in A}.$$

This is the $\{0, 1\}$ -valued charge corresponding to $\mathcal{U}$: $\ell(A) = 1$ iff $A \in \mathcal{U}$. Since $\mathcal{U}$ is nonprincipal, $\ell(\{n\}) = 0$ for every n , yet $\ell(\mathbb{N}) = 1$. Thus the sets $A_k = \{k, k+1, \dots\}$ satisfy $A_0 \supseteq A_1 \supseteq \dots$ and $\bigcap_k A_k = \emptyset$, yet $\ell(A_k) = 1$ for all k (since $A_k \in \mathcal{U}$ for all k , as $\mathcal{U}$ is nonprincipal). Hence ℓ is not σ -additive.

If the ultraproduct's charge were σ -additive, its restriction to the diagonal copy of $\mathcal{P}(\mathbb{N})$ would be σ -additive; but this restriction is ℓ , which is not. Hence the ultraproduct does not satisfy σ -additivity, contradicting Łoś's theorem applied to the assumption that each $(\mathcal{P}(\mathbb{N}), \delta_n) \models T$. $\square$

Remark 3. The ultraproduct $\prod_{\mathcal{U}}(\mathcal{P}(\mathbb{N}), \delta_n)$ is an $\mathcal{L}_{\text{BA}, \mu}$ -structure in its own right; ℓ is the charge induced on the diagonal copy of $\mathcal{P}(\mathbb{N})$ inside it. The diagonal copy is isomorphic to $\mathcal{P}(\mathbb{N})$ as a Boolean algebra, and ℓ is the $\{0, 1\}$ -valued ultrafilter charge on it: a purely finitely additive probability that is not σ -additive.

The proof exhibits a purely finitely additive charge — the ultrafilter charge on $\mathcal{P}(\mathbb{N})$ — as a $\mathcal{U}$ -ultralimit of σ -additive probabilities. This shows that the class of σ -additive structures is not closed under ultraproducts, hence not first-order axiomatizable. A natural follow-up question is which purely finitely additive charges on a Boolean algebra arise as pointwise ultralimits of σ -additive probabilities on the same algebra; the answer depends sensitively on the algebra's structural properties and remains open in the non- σ -complete non-atomic case.

References

- [1] Ronald Fagin, Joseph Y. Halpern, and Nimrod Megiddo. A logic for reasoning about probabilities. *Information and Computation*, 87(1–2):78–128, 1990.
- [2] Douglas N. Hoover. Probability logic. *Annals of Mathematical Logic*, 14(3):287–313, 1978.
- [3] H. Jerome Keisler. Probability quantifiers. In Jon Barwise and Solomon Feferman, editors, *Model-Theoretic Logics*, pages 509–556. Springer, New York, 1985.
- [4] Jerzy Łoś. Quelques remarques, théorèmes et problèmes sur les classes définissables d’algèbres. In *Mathematical Interpretation of Formal Systems*, pages 98–113. North-Holland, 1955.